

A crude oil price forecasting framework based on Constraint Guarantee and Pareto Fronts Shrinking Strategy

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ABSTRACT

Accurate forecasting of crude oil prices is essential for making informed energy policy decisions and ensuring energy security. However, crude oil price forecasting is inherently challenging due to the volatile, nonlinear, and complex nature of the market. While ensemble learning approaches have shown promise in enhancing forecasting accuracy, many existing models rely on multi-objective optimization techniques that generate a Pareto frontier of optimal solutions, often making it difficult to select the best solution for practical application. This issue is exacerbated by the fact that some Pareto-optimal solutions are not suitable for real-world decision-making, leading to inefficiencies in model performance. To address these limitations, this research proposes a novel ensemble learning framework that incorporates a Constraint Guarantee Strategy (CGS) and a Pareto Front Shrinking Strategy (PFSS) to enhance both the accuracy and stability of crude oil price forecasting models. The CGS filters out inferior solutions during the optimization process, ensuring that the ensemble model outperforms individual models in terms of forecasting accuracy. The PFSS helps decision-makers select the most relevant solutions from the Pareto frontier by balancing trade-offs between objectives and narrowing down the set of solutions. Our framework is evaluated on three widely used datasets: Brent, WTI, and Dubai crude oil prices, and compared with state-of-the-art models from both the general time-series forecasting domain and crude oil price forecasting. It improves prediction accuracy by approximately 23.2% on the Brent dataset, 4.0% on the WTI dataset, and 21.7% on the Dubai dataset, based on improvements in MAPE. Ablation studies confirm the effectiveness of each component. The discussion further emphasizes the practical applicability and robustness of the framework, confirming its potential for real-world crude oil price forecasting.

1. Introduction

1.1. Background and motivation

Accurate forecasting of crude oil prices is critical for formulating effective energy policies, as it directly influences strategic decisions related to energy production, consumption, and environmental sustainability [1]. However, the volatility of crude oil prices, driven by a complex mix of factors such as geopolitical events, economic fluctuations, and market speculation, makes the forecasting process particularly challenging [2,3]. Additionally, the non-linearity, non-stationarity, and randomness inherent in crude oil price movements further amplify these challenges, requiring advanced forecasting methods that can effectively capture these dynamic and multifaceted influences.

To enhance forecasting accuracy, researchers have turned to ensemble learning approaches, which combine multiple models to leverage their complementary strengths. Multi-objective optimization algorithms, in particular, have been widely used to assign appropriate weights to the individual models' outputs, thereby improving both accuracy and stability. However, the use of multi-objective optimization often leads to the generation of a set of Pareto-optimal solutions rather than a single, definitive solution [4,5]. While these solutions are theoretically optimal, many are not practically applicable [6], leading to inefficiencies in the ensemble. Selecting inferior solutions from this set can degrade the overall performance, undermining the benefits of combining models. To address this challenge, we propose a novel ensemble framework incorporating a constraint guarantee and pareto frontier shrinking strategy, ensuring that the integrated model outperforms the best individual model. This approach aims to improve

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both the precision of the forecasts and the decision-making process, while also enhancing the robustness and generalizability of crude oil price prediction models [7].

1.2. Related works

Crude oil price forecasting (COPF) frameworks have made significant progress over recent decades [8]. The methodologies are broadly classified into three main categories: classical statistical models, artificial intelligence and machine learning models, and hybrid models [9, 10]. Initially, forecasters relied on linear regression, GARCH models, and ARIMA models [11,12], which provided a foundational framework for understanding price dynamics within the crude oil market. These methods, primarily grounded in historical price, were the mainstay of early forecasting efforts. For example, Ying Xiang et al. applied the ARIMA model to forecast Brent crude oil prices, demonstrating that the ARIMA(1,1,1) model yielded commendable predictive accuracy for short-term international oil prices [13]. Michael Ye et al. enhanced model fit and predictive capability for short-term WTI crude oil prices by incorporating low and high inventory states within a single-equation model [14]. However, these models struggle to capture the nonlinear dynamics of crude oil prices and often exhibit lower accuracy [15]. As the crude oil market's complexity and nonlinear dynamics became increasingly apparent, the limitations of traditional statistical approaches prompted a shift towards AI methodologies [16]. Herrera et al. evaluated the forecast performance of traditional econometric models against machine learning methods (Neural Networks and Random Forests) for major energy commodities using monthly prices from the IMF, finding that machine learning techniques not only outperform traditional methods in terms of RMSE and MAPE but also demonstrate a unique advantage in predicting market turning points [17]. Methods like artificial neural networks (ANNs) [18] and recurrent neural networks have become prominent for their proficiency in handling complex data structures, significantly enhancing the capture of nonlinear patterns in crude oil prices [19–21]. Bin Wang et al. developed a novel random deep bidirectional gated recurrent unit neural network with a random inheritance formula, enhancing the accuracy of forecasting international crude oil futures prices [22]. While single artificial intelligence models have significantly improved the accuracy of crude oil price forecasting, they exhibit considerable performance variation across different datasets and are sensitive to parameter settings, indicating substantial room for improvement.

In recent years, researchers have increasingly focused on hybrid models to further improve crude oil price forecasting accuracy by combining multiple models, either vertically or horizontally. Vertical integration refers to combining different types of models within a single framework in a sequential manner, where each model operates at a different stage [23–25]. For example, Gong et al. proposed a combined model that integrates secondary decomposition with deep learning techniques for more accurate predictions of crude oil futures price and volatility [26]. On the other hand, horizontal integration involves combining predictions from multiple models of the same type or different types at the same level, which is commonly known as ensemble learning. Ensemble learning aggregates the predictions of individual models to enhance the overall forecasting accuracy. Some researchers have employed single-objective optimization algorithms to aggregate model results by minimizing the error level or adjusting each model's weight before linearly combining the predictions to produce a final output. For example, Wang et al. used a single-objective optimization algorithm to integrate five different models, improving the accuracy of crude oil price forecasting [27]. Liu et al. proposed a time-varying weight combined model, incorporating multiple forecasting techniques and dynamically adjusting model weights based on Shannon entropy, further improving prediction accuracy and robustness during uncertain times [28]. However, this approach, which primarily focuses on accuracy, often overlooks model stability, a crucial factor

for reliable forecasting. To address this, some researchers have introduced multi-objective optimization-based ensemble methods, which aim to maximize both accuracy and stability. For example, Jiang et al. proposed a Decomposition-Selection-Ensemble system using a multi-objective Chaos Game Optimization algorithm, which improves both accuracy and stability in energy futures price forecasting [29].

Despite these advancements, several challenges persist. Multi-objective optimization algorithms generate a set of optimal solutions, referred to as the Pareto frontier, instead of a single optimal solution. If the solution set contains ineffective or irrelevant solutions, it can lead to suboptimal ensemble results. Furthermore, current research often selects solutions randomly from the Pareto frontier, with limited focus on methods for objectively identifying the best solution. To ensure that ensemble outcomes consistently outperform individual models and to guide decision-makers in the selection process, we propose a novel ensemble framework that integrates a Constraint Guarantee Strategy (CGS) and a Pareto Front Shrinking Strategy (PFSS). The CGS constrains the Pareto frontier, filtering out solutions that lack practical significance, ensuring that the ensemble results are superior to those of any individual model. Meanwhile, the PFSS reduces the size of the Pareto frontier by balancing trade-offs between objectives, helping decision-makers choose the optimal solution from a smaller, more meaningful set. To the best of our knowledge, we are the first to incorporate CGS and PFSS into ensemble learning based on multi-objective optimization for crude oil price forecasting, resulting in enhanced accuracy and stability in the forecasts. Table 1 contrasts our work with existing literature. Table 2 evaluates some typical existing crude oil price forecasting models.

1.3. Our contributions

This research presents a novel ensemble learning framework designed to improve the accuracy, stability, and applicability of crude oil price forecasts. The framework is composed of three key modules: data preprocessing, ensemble learning via a multi-objective optimizer, and a customized PFSS. The major innovations of our work lie in the introduction of the Constraint Guarantee Strategy (CGS) and the Pareto Front Shrinking Strategy (PFSS). The CGS penalizes solutions that perform worse than the best forecasting results in the model pool midway through the optimization process, significantly enhancing the quality of the Pareto frontier and reducing inefficient solutions. Meanwhile, the PFSS is designed to reflect both the trade-offs between different solutions and the trade-offs between objective functions, enabling decision-makers to objectively and flexibly select a small set of solutions most relevant to their needs, even in the absence of prior knowledge. By integrating CGS and PFSS, our approach not only improves forecast precision and stability but also enhances the practical applicability of the solutions selected by decision-makers.

The major contributions of this research can be summarized as follows:

- **A ensemble learning framework that simultaneously considers accuracy and stability:** This framework integrates multiple models using a multi-objective optimization approach, simultaneously improving prediction accuracy and stability. By incorporating the Constraint Guarantee Strategy and Pareto Front Shrinking Strategy during the integration process, the framework further enhances accuracy, demonstrating robust performance across different datasets.
- **Evolving FTTA to HMOFTTA, markedly surpassing classical multi-objective algorithms:** We modified the Football Team Training Algorithm (FTTA), originally a single-objective optimization algorithm, into Hyper-dominance-based Multi-objective Football Team Training Algorithm (HMOFTTA) to better suit our multi-objective ensemble framework. By introducing improvements, we significantly enhanced its multi-objective optimization performance, ensuring better convergence and adaptability for our framework.

Table 1
Summary of the related works.

Related Works		Feature Eng.	ML	Ensemble	Multi-obj. opt.	PF Shrinking
Classical Statistical models	[12,13]	✗	✗	✗	✗	✗
	[14]	✓	✗	✗	✗	✗
	[11]	✗	✗	✓	✗	✗
AI Based Model	[30]	✓	✓	✓	✗	✗
	[11,22]	✗	✓	✗	✗	✗
Hybrid Models	[23–25]	✗	✓	✓	✗	✗
	[31]	✓	✓	✓	✗	✗
	[32,33]	✗	✓	✓	✓	✗
	Our work	✓	✓	✓	✓	✓

Table 2
The evaluation of existing crude oil price forecasting models.

Models	Reference	Benchmarks	Description	Advantage	Disadvantage
Statistical Models	[13]	Brent	The ARIMA (1,1,1) model has good performance and can be used for short-term crude oil price forecasting.	High computational efficiency, extensively used for processing linear time-series data.	Limited accuracy, struggles with nonlinear dynamics of crude oil prices.
	[14]	WTI	Nonlinear inventory-price relationship significantly improves short-run crude oil price forecasting.		
AI-Based Models	[22]	Brent WTI	A deep bidirectional gated recurrent unit neural network, enhancing the accuracy of forecasting crude oil futures prices.	High predictive accuracy, can handle complex nonlinear data structures.	Tends to overfit noisy data, sensitive to parameter changes, poor robustness.
	[34]	WTI	LSTM-based model for predicting WTI crude oil prices, outperforming existing models.		
Hybrid Models	[28]	Brent WTI	Dynamically adjusts model weights based on Shannon entropy for improved forecasting accuracy and robustness.	Hybrid models combine various forecasting techniques, improving accuracy and robustness. By integrating multiple models, they reduce prediction errors and balance both accuracy and stability, offering more reliable forecasts.	Ensemble methods based on multi-objective optimization can struggle with handling the Pareto frontier, leading to suboptimal or irrelevant solutions. Selecting the best solution from multiple optimal results can also be arbitrary, impacting model effectiveness.
	[26]	Brent WTI	A novel deep learning model combining secondary decomposition with BiGRU-Attention-CNN for price forecasting, outperforming other methods.		
	[24]	Brent	Combines VMD, PSR, CNN-BiLSTM to predict crude oil prices, achieving superior performance.		
	[27]	Brent	Integrates five different models using an optimization algorithm for improved forecast accuracy.		

- **The CGS for enhancing the quality of Pareto front solutions:** We introduced the CGS, ensuring the forecasting performance of the ensemble model markedly outperforms that of any single model within the pool. The CGS commences penalizing solutions with inferior predictive results compared to the optimal outcomes in the model pool halfway through the optimization algorithm iterations. This approach not only prevents the optimization algorithm from searching in inferior solution regions but also significantly enhances the quality of the Pareto front optimal solutions.
- **A Customized PFSS for objectively selecting Pareto front solutions:** Despite achieving a high-quality Pareto front, objectively selecting the best solution from the frontier remains a challenge. The PFSS refines the Pareto frontier by balancing trade-offs and helps decision-makers in selecting the most relevant solutions, ensuring a more efficient and informed decision-making process.

The remainder of the research is structured as follows: Section 2 introduces the theoretical background of the algorithms utilized in the proposed framework. Section 3 covers the data selection, parameter design, and evaluation metrics. Section 4 demonstrates the significant advantages of each module within the framework through five sets of

experiments. Section 5 discusses the practical applicability and robustness of the framework. Section 6 concludes the research with remarks and identifies potential future research directions.

2. Methodology

2.1. Data preprocessing module

2.1.1. Successive variational mode decomposition (SVMD)

Variational Mode Decomposition (VMD) is adept at decomposing a signal into sub-signals with specific central frequencies. However, its efficacy diminishes when the number of modes is unknown, leading to parameter sensitivity and computational burdens [35]. SVMD addresses these limitations by successively extracting modes without the need to predetermine their count [36]. The decomposition process is governed by the following optimization problem:

$$\min_{\{u_i\}, \{\Omega_i\}} \sum_{i=1}^I \left\| \partial_t \left[\left(\rho(t) + \frac{j}{\pi t} \right) * u_i(t) \right] e^{-j\Omega_i t} \right\|_2^2 \quad (1)$$

where the objective is to minimize the bandwidths of the modes $u_i(t)$, each associated with a center frequency Ω_i while ensuring that the sum

of these modes reconstructs the original signal $s(t)$:

$$\text{such that, } \sum_{i=1}^I u_i(t) = s(t) \quad (2)$$

The SVMd method employs an augmented Lagrangian to guide the decomposition, which facilitates effective constraint management and targets desired signal properties by integrating a quadratic penalty and a Lagrangian multiplier and facilitates the optimization through dual ascent [37].

$$\begin{aligned} G(\{u_i\}, \{\Omega_i\}, \mu) = & \beta \sum_{i=1}^I \left\| \partial_t \left[\left(\rho(t) + \frac{j}{\pi t} \right) * u_i(t) \right] e^{-j\Omega_i t} \right\|_2^2 \\ & + \left\| s(t) - \sum_{i=1}^I u_i(t) \right\|_2^2 + \left\langle \mu(t), s(t) - \sum_{i=1}^I u_i(t) \right\rangle \\ & + \left\| s(t) - \left(u_I(t) + s_u(t) + \sum_{i=1}^{I-1} u_i(t) \right) \right\|_2^2 \\ & + \left\langle \mu(t), s(t) - \left(u_I(t) + s_u(t) + \sum_{i=1}^{I-1} u_i(t) \right) \right\rangle. \end{aligned} \quad (3)$$

With each iteration, SVMd refines the mode functions and their associated frequencies, progressively leading to an accurate and computationally efficient decomposition of the original signal.

2.1.2. Hampel identifier (HI)

HI is an efficient method for detecting outliers within a data series $X = \{x_1, x_2, \dots, x_n\}$ [32]. It employs a sliding window to calculate the local median m_i and modified standard deviation σ_i , using:

$$m_i = \text{median}\{x_{i-k}, \dots, x_{i+k}\} \quad (4)$$

$$\sigma_i = k' \cdot \text{median}\{|x_{i-k} - m_i|, \dots, |x_{i+k} - m_i|\} \quad (5)$$

with $k' \approx 1.4826$.

Outliers are identified if they deviate from the local median by more than n_0 times σ_i :

$$y_i = \begin{cases} x_i, & \text{if } |x_i - m_i| \leq n_0 \sigma_i \\ \text{outlier}, & \text{if } |x_i - m_i| > n_0 \sigma_i \end{cases} \quad (6)$$

where n_0 is typically set to 3.

2.1.3. Relieff algorithm for feature selection

The Relieff algorithm is an enhancement of the Relief algorithm, designed for feature selection in datasets with complex feature interactions. Relieff is adapted for multi-class datasets and is robust to noisy data [38]. It operates by randomly selecting instances and updating feature weights based on the differences between the chosen instance and its nearest neighbors within the same class (hits) and from other classes (misses).

The formula for updating the weight of a feature A in Relieff is given by:

$$W(A) = W(A) - \sum_{j=1}^k \frac{\text{diff}(A, R, H_j)}{mk} + \sum_{C \neq \text{class}(R)} \left[\frac{P(C)}{1 - P(\text{class}(R))} \sum_{j=1}^k \frac{\text{diff}(A, R, M_j^{(C)})}{mk} \right] \quad (7)$$

where $W(A)$ represents the weight of the attribute A , R is a randomly selected instance, H_j denotes the nearest hit, $M_j^{(C)}$ denotes the nearest miss from a different class C , k is the number of nearest neighbors, and m is the number of instances sampled. The difference function $\text{diff}(A, R_1, R_2)$ is defined as follows:

Table 3

Pseudocode for the Constraint Guarantee Strategy (CGS).

Algorithm: Constraint Guarantee Strategy (CGS)

Input: Model pool $\mathcal{M} = \{M_1, M_2, \dots, M_n\}$
 where each M_i is a single model, $MaxIter$

Output: Refined solution set S_{refined}

Objective Function: For a given solution S_j , define:

$$o(1)_{S_j} = \frac{\text{mean}(|Y_i(S_j) - \text{actual}|)}{\text{actual}} \times 100$$

$$o(2)_{S_j} = \text{std}(Y_i(S_j) - \text{actual})$$

Initialization:
 Set $MAPE^* = \min_{M \in \mathcal{M}} \{MAPE(M)\}$
 Set $std^* = \min_{M \in \mathcal{M}} \{std(M)\}$
 Initialize iteration count $n = 0$
 Generate initial solution set S

while $n < MaxIter$ **do**
 Update solution set S using optimization algorithm
 $n = n + 1$
if $n > 0.5 \times MaxIter$ **then**
 for each solution S_j in S **do**
 Calculate $o(1)_{S_j}$ and $o(2)_{S_j}$
 if $o(1)_{S_j} > MAPE^*$ **or** $o(2)_{S_j} > std^*$ **then**
 Penalize S_j by setting $F(S_j) = (\infty, \infty)$
 end if
 end for
end if
end while
Return S_{refined}

$$\text{diff}(A, R_1, R_2) = \begin{cases} \frac{|R_1[A] - R_2[A]|}{\max(A) - \min(A)} & \text{if } A \text{ is Continuous} \\ 0 & \text{if } A \text{ is Discrete and } R_1[A] = R_2[A] \\ 1 & \text{if } A \text{ is Discrete and } R_1[A] \neq R_2[A] \end{cases} \quad (8)$$

Relieff advances the Relief algorithm by accommodating multi-class scenarios, handling noisy and incomplete data effectively.

2.2. Multi-objective optimization ensemble learning module based on constraint guarantee

In this subsection, we expound on the design of the Multi-Objective Optimization Ensemble Learning module, anchored by the Constraint Guarantee Strategy (CGS).

The proposed ensemble learning method is based on a multi-objective optimization algorithm that simultaneously considers multiple objectives. In our framework, we focus on both prediction accuracy and stability. To represent accuracy, we use the Mean Absolute Percentage Error (MAPE), while the standard deviation (STD) of the error serves as a measure of stability. By minimizing MAPE and STD, we effectively maximize both accuracy and stability. The objective functions for this multi-objective optimization are given in Eq. (9).

$$\min_x \begin{cases} Fobj(1) = \text{mean} \left(\frac{|Y_i - \text{actual}|}{\text{actual}} \right) \times 100 \\ Fobj(2) = \text{std}(Y_i - \text{actual}) \end{cases} \quad s.t. \quad \begin{cases} Fobj(1) < MAPE^* \\ Fobj(2) < Std^* \end{cases} \quad (9)$$

where $Y_i = x^{(1)}MLP + x^{(2)}CNN + x^{(3)}BiLSTM + x^{(4)}BiGRU + x^{(5)}SVM$.

However, multi-objective optimization typically results in a set of optimal solutions (i.e., a Pareto frontier) rather than a single, unique solution. In practice, we observe that many of the solutions generated during optimization do not offer any meaningful improvement over the best model in the pool, i.e., their MAPE and STD are worse than those of the best individual model in the ensemble. This leads to inefficiency in the optimization process. To address this issue, we introduce the Constraint Guarantee Strategy (CGS), which filters out these ineffective solutions. The CGS ensures that the solutions on the Pareto frontier improve upon the best model in terms of both MAPE and STD, thus guaranteeing that the ensemble's performance exceeds that of any

single model in the pool. This constraint is applied only in the latter half of the optimization process. During the early stages, we aim to obtain diverse solutions, which helps explore the solution space. In the latter stages, however, we focus on refining the solutions to obtain higher-quality, more meaningful results. The detailed process of CGS is outlined in the pseudocode in Table 3.

Once the objective functions are defined, we must determine which models to integrate and what optimization algorithm to use. Traditional single AI models are highly sensitive to both parameters and data, and their performance can vary significantly across different datasets. To leverage the strengths of various models and ensure robust performance across a wide range of datasets, we integrate multiple models, including deep learning models (such as CNN, BiLSTM, BiGRU) and shallow models (such as SVM and MLP). This diversified model pool ensures that the ensemble framework performs well across different forecasting tasks.

To further enhance the optimization of the defined objective functions, we modify a powerful single-objective optimization algorithm, the Football Team Training Algorithm (FTTA), and extend it to a multi-objective version, called HMOFTTA. This modification enables the algorithm to effectively handle multi-objective problems and determine the optimal weights for each model in the ensemble, ensuring that the final integrated model achieves both high accuracy and stability.

The methodology of the forecasting model and the improved optimization algorithm is shown below:

2.2.1. Model pool

In the model pool, we selected both deep learning models (CNN [39], BiLSTM [40,41], and BiGRU [42]) and shallow models (SVM [43] and MLP [6]). The reason for including these models is that they have different characteristics and can leverage the strengths of various modeling approaches. Deep learning models, such as CNN, BiLSTM, and BiGRU, are well-suited for capturing complex patterns and long-range dependencies in time-series data. On the other hand, shallow models like SVM and MLP are more efficient in handling smaller datasets and often excel in situations where the relationship between input and output is less complex. The performance of these models can vary significantly across different datasets. Therefore, by integrating different types of models, we aim to combine their individual strengths and achieve a more robust performance across diverse datasets.

2.2.2. Hyper-dominance-based Multi-objective Football Team Training Algorithm (HMOFTTA)

Football Team Training Algorithm (FTTA):

The FTTA is a metaheuristic optimization framework inspired by the strategic and adaptive training approaches found in football teams [16]. The FTTA algorithm comprises three main stages: collective training, group training, and individual extra training. Each stage is designed to enhance the solutions iteratively:

Collective Training: In the Collective Training phase of the FTTA algorithm, solutions are assessed and then categorized as Followers, Discoverers, Thinkers, or Volatilities to guide their improvement strategies.

$$F_{ij}^{k,new} = \begin{cases} F_{ij}^{k,old} + \text{rand} \times (F_{Best,j}^k - F_{ij}^{k,old}), & \text{for Followers} \\ F_{ij}^{k,old} + \text{rand}_1 \times (F_{Best,j}^k - F_{ij}^{k,old}) - \text{rand}_2 \times (F_{Worst,j}^k - F_{ij}^{k,old}), & \text{for Discoverers} \\ F_{ij}^{k,old} + \text{rand} \times (F_{Best,j}^k - F_{Worst,j}^k), & \text{for Thinkers} \\ F_{ij}^{k,old} \times (1 + t(k)), & \text{for Volatilities} \end{cases} \quad (10)$$

where $F_{ij}^{k,old}$ is the prior fitness value, $F_{ij}^{k,new}$ the updated fitness value for the i -th player in dimension j , $F_{Best,j}^k$ the fitness value of the best player in dimension j at iteration k , $F_{Worst,j}^k$ the fitness of the worst

player in dimension j at iteration k , and rand , rand_1 , rand_2 are random numbers. $t(k)$ is a random number with t -distribution which its degree of freedom is the current number of iterations.

Group Training: After completing the collective training, the football training process reaches the group training stage. Players are divided into four categories based on their characteristics: Strikers, Midfielders, Defenders, and Goalkeepers, using the MGEM (MixGaussEM) clustering algorithm. In group training, players improve through three primary learning mechanisms and one error mechanism, selected randomly and applied based on specific probabilities.

Optimal Learning: A player learns from the best player in the group with a probability P_{study} :

$$F_{i,j}^{\text{team},\text{new}} = \begin{cases} F_{Best,j}^{\text{team},k} & \text{if } \text{rand} \leq P_{\text{study}}, \\ F_{i,j}^{\text{team},k} & \text{otherwise.} \end{cases} \quad (11)$$

Random Learning: A player learns from a random player in the group with a probability P_{study} :

$$F_{i,j}^{\text{team},\text{new}} = \begin{cases} F_{\text{Random},j}^{\text{team},k} & \text{if } \text{rand} \leq P_{\text{study}}, \\ F_{i,j}^{\text{team},k} & \text{otherwise.} \end{cases} \quad (12)$$

Random Communication: Players exchange abilities with a probability P_{comm} :

$$F_{i,j}^{\text{team},\text{new}} = \begin{cases} F_{\text{Random},j}^{\text{team},k} \times (1 + \text{randn}) & \text{if } \text{rand} \leq P_{\text{comm}}, \\ F_{i,j}^{\text{team},k} & \text{otherwise.} \end{cases} \quad (13)$$

Random Error: Players learn incorrect information with a small probability P_{error} :

$$F_{i,j}^{\text{team},\text{new}} = \begin{cases} F_{\text{Random1,Random2},j}^{\text{team},k} & \text{if } \text{rand} \leq P_{\text{error}}, \\ F_{i,j}^{\text{team},k} & \text{otherwise.} \end{cases} \quad (14)$$

In these equations, $F_{i,j}^{\text{team},\text{new}}$ represents the updated fitness value of the i -th player in the j -th dimension of team l after iteration k . The variables rand and randn are uniformly and normally distributed random numbers, respectively. The probabilities P_{study} , P_{comm} , and P_{error} control the likelihood of each learning mechanism.

Individual Extra Training: The Individual Extra Training phase in FTFA capitalizes on refining the top performer's capabilities post group training. It employs a mixture of Cauchy and Gaussian distributions to adjust the best player's fitness, as detailed in the equation below:

$$F_{Best}^{k,new} = F_{Best}^{k,old} \times \left(1 + \left(1 - \frac{1}{k} \right) \times \text{Gauss} + \frac{1}{k} \times \text{Cauchy} \right) \quad (15)$$

This equation represents the adaptive training methodology where $F_{Best}^{k,new}$ and $F_{Best}^{k,old}$ are the new and old fitness values of the best player, k denotes the iteration count, and Gauss and Cauchy are the stochastic variables from the respective distributions. This strategic update not only boosts the best player's performance but also sets a higher standard within the team, progressively shifting the focus from a broad exploration to a more targeted exploitation as the algorithm iterates.

Hyper-dominance-based Multi-objective Football Team Training Algorithm:

Building upon the foundations of the FTFA, we present a novel approach known as the Hyper-dominance-based HMOFTTA. This enhanced algorithm introduces significant improvements in initialization and optimization phases to address multi-objective optimization challenges more effectively.

Improvement I: Initialization of the Population.

In the FTFA, initializing the population with random numbers leads to an uneven distribution of the initial population, which may result

in convergence to local optima. To address it, the initial population is generated using a Halton low-discrepancy sequence to ensure a uniform distribution of solutions. Additionally, opposite solutions are created using an opposition-based strategy to increase the diversity of the initial population. Subsequently, non-dominated sorting and crowding distance calculations are used to rank individuals, and the individuals with the highest rankings equal to the initial population size are selected for the next generation.

Halton Sequence: The Halton sequence is an extension of the Van der Corput sequence to multiple dimensions, used for generating values that are evenly distributed in a unit hypercube. It is defined by the radical inversion function and the use of different bases for each dimension, ensuring low discrepancy for quasi-random sampling in multi-dimensional spaces.

Given a base b , the radical inversion function Φ_b of a non-negative integer i is given by:

$$i = \sum_{l=0}^{M-1} a_l(i)b^l, \quad \Phi_b(i) = \sum_{l=0}^{M-1} \frac{a_l(i)}{b^{l+1}} \quad (16)$$

where $a_l(i)$ is the l th digit of i in base b . The Van der Corput sequence is generated by applying the radical inversion function to the sequence of natural numbers.

For higher dimensions, the Halton sequence uses distinct prime bases $b_1, \dots, b_n$ for each dimension. The sequence in n -dimensions is constructed as follows:

$$X_i = (\Phi_{b_1}(i), \dots, \Phi_{b_n}(i)) \quad (17)$$

where each $\Phi_{b_j}(i)$ is the radical inversion of i in base b_j .

Opposition-Based Strategy: After obtaining the Halton sequence, an opposition-based strategy is employed to generate opposing solutions. The combined set of Halton and opposing solutions undergo non-dominated sorting and crowding distance calculations. The top solutions based on these criteria are selected as the initial population pool.

The opposing solutions are computed as follows:

$$X' = \text{rand}() \cdot (Lb + Ub - X) \quad (18)$$

where X' is the opposing solution, $\text{rand}()$ is a random number generator function, Lb , and Ub are the lower and upper bounds of the search space, respectively, and X is the original solution obtained from the Halton sequence. The initial population is then formed by selecting the foremost solutions according to non-dominance and spatial distribution.

Improvement II: Hyper-dominance Degree. To enhance the Pareto front and mitigate the issue of solutions that are exceptional in one objective but notably deficient in others, the hyper-dominance degree is utilized to refine the selection process, which assesses solution convergence, thus discriminating against those with extreme trade-offs in multi-objective performance. This refinement helps to promote a more balanced set of solutions [31].

Hyper-dominance degree: For a solution set $S = \{x_1, x_2, \dots, x_n\}$, the hyper-dominance degree of the i -th solution hd_i is defined as:

$$hd_i = \min(H(x_i, x_1), H(x_i, x_2), \dots, H(x_i, x_n)) \quad (19)$$

where $H(x_i, x_j)$ is given by:

$$H(x_i, x_j) = \begin{cases} \infty, & \text{if } \max(x_i - x_j) \leq 0, \\ -\frac{\max(x_i - x_j)}{\min(x_i - x_j)}, & \text{if } \max(x_i - x_j) > 0. \end{cases} \quad (20)$$

where $H(x_i, x_j)$ represents the hyper-dominance degree of solution x_i against x_j , quantifying the extent of dominance or non-dominance between solutions. We define a linearly adaptive tolerance parameter (T) within our iterative process. After each iteration, we calculate the hyper-dominance degree (hd_i) for solutions, excluding those where

Table 4

Pseudocode for the HMOFTTA Algorithm.

Algorithm: HMOFTTA
Input: $P, \text{MaxIts}, \text{MaxT}, Lb, Ub$ $P_{\text{halton}} \leftarrow \text{HaltonSequence}()$ $P_{\text{opp}} \leftarrow \text{Opposition_Strategy}(P_{\text{halton}}, Lb, Ub)$ $P_0 \leftarrow \text{Merge}(P_{\text{halton}}, P_{\text{opp}})$ $P_0 \leftarrow \text{NonDominatedSort\&CrowdingDistance}(P_0)$ $P_0 \leftarrow \text{SelectTop}(P_0, \text{pop})$ for $j = 1$ to MaxIts do Apply CollectiveTraining, GroupTraining, and IndividualExtraTraining $T_j \leftarrow \frac{j}{\text{MaxIts}} \times \text{MaxT}$ for each $x_i \in P_0$ do $hd_i \leftarrow \min(\{H(x_i, x_j) \forall x_j \in P_0\})$ $H(x_i, x_j) \leftarrow \begin{cases} \infty & \text{if } \max(x_i - x_j) \leq 0 \\ -\frac{\max(x_i - x_j)}{\min(x_i - x_j)} & \text{if } \max(x_i - x_j) > 0 \end{cases}$ end for $P_0 \leftarrow P_0 \setminus \{x_i hd_i < T_j\}$ end for

$hd_i < T$. This selective criterion aims to curate a set of solutions with enhanced quality, ensuring a harmonious balance between the convergence and diversity of the solution set.

Tolerance Adjusting Strategy: The tolerance T plays a pivotal role in applying selection pressure. By setting a variable tolerance, which adapts throughout the optimization iterations, we can preserve diversity at the initial stages and gradually apply stronger convergence pressure as the process continues. This dynamic approach is formalized by the following linear growth model:

$$T_j = \text{MaxT} \cdot \frac{j}{\text{MaxIts}} \quad (21)$$

where T_j is the tolerance value at iteration j , MaxT is the predetermined maximum value for tolerance, and MaxIts represents the maximum number of iterations. The linear growth of T from 0 to MaxT allows for controlled selection pressure that gently increases, ensuring a balance between diversity at the early stages of optimization and convergence as the process nears completion. Table 4 illustrates the process of HMOFTTA.

2.3. Pareto frontier shrinking strategy module (PFSS)

Although we have filtered out some solutions that are meaningless for the ensemble through constraints, the final choice still involves selecting just one solution. How to objectively choose a solution set from a series of Pareto fronts is crucial. Traditional methods of selecting solutions on the Pareto front often resort to random sampling or exclusive focus on knee points. While knee points are excellent for highlighting critical trade-offs between pairs of solutions, they may not universally indicate the best overall system performance. Similarly, regions defined by competition efficiency focus on the trade-offs at the extreme values of the objective functions rather than across the entire Pareto front. In this part, we introduce the Customized PFSS, a technique that synergizes the distinctive qualities of both the Knee Point Identification Using Knee Point Identification Based on Trade-off Utility (KPITU) and the Shrinking Method to Quantify Competition (SMQC). The design objective of PFSS is to contract the Pareto frontier into a highly efficient selection area known as the Final Shrinking Region (FSR). Within this narrow region, decision-makers can flexibly choose Pareto solutions, achieving highly precise predictive outcomes.

The FSR strategy surpasses these methods by intersecting the knee point region, which emphasizes inter-solution trade-offs, with the competition efficiency region, which focuses on the trade-offs related to the objectives' extreme values.

2.3.1. Knee point identification based on trade-off utility (KPITU)

Knee points represent pivotal solutions where an improvement in one objective leads to a significant degradation in another. These are critical for DMs in multi-objective optimization. The KPITU algorithm identifies these knee points within a set of non-dominated solutions, leveraging knee dominance to discern the most preferred solutions systematically [44].

Trade-off Utility: The trade-off utility for two non-dominated solutions x^1 and x^2 is:

$$U(x^1, x^2) = G(x^1, x^2) + D(x^1, x^2), \quad (22)$$

where G and D represent the net gain and deterioration and are defined as:

$$G(x^1, x^2) = \sum_{i=1}^m \left(\frac{f_i(x^1) - f_i(x^2)}{z_i^{\max} - z_i^{\min}} \right)^+, \quad (23)$$

$$D(x^1, x^2) = \sum_{i=1}^m \left(\frac{f_i(x^1) - f_i(x^2)}{z_i^{\max} - z_i^{\min}} \right)^-, \quad (24)$$

with $[x]^+ = \min\{0, x\}$ and $[x]^- = \max\{x, 0\}$.

Knee-Dominance and Local Knee Points: A solution x^1 knee-dominates x^2 , written as $x^1 \prec_k x^2$, if $U(x^1, x^2) < 0$. A solution x^* is a local knee point if no other solution x' in the same subregion provides better trade-off utility. The knee-dominance is then an asymmetric and transitive relation that serves to identify these knee points.

Algorithmic Procedure: KPITU progresses by:

1. Partitioning the objective space into subregions using weight vectors.
2. Associating each solution with a subregion.
3. Identifying local knee points by evaluating trade-off utility.

For a set of local knee points K , the accumulated trade-off utility for x_k^i is calculated by summing up the trade-off utilities against all other points in K :

$$\mathcal{K}(x_k^i) = \sum_{j=1, j \neq i}^{|K|} U(x_k^i, x_k^j). \quad (25)$$

The optimal knee point is then selected by maximizing $\mathcal{K}$, signifying the best trade-off among all considered objectives.

2.3.2. Shrinking Method to Quantify Competition (SMQC)

Competition efficiency quantifies the benefit to loss ratio when transitioning between states where objectives reach their maximal values [45].

$$\eta_{12} = \frac{\hat{Z}_1^{\max} - Z_1^B}{\hat{Z}_1^{\max}} \times \frac{\hat{Z}_2^{\max} - Z_2^A}{\hat{Z}_2^{\max}} \quad (26)$$

$$\eta_{21} = \frac{\hat{Z}_2^{\max} - Z_2^A}{\hat{Z}_2^{\max}} \times \frac{\hat{Z}_1^{\max} - Z_1^B}{\hat{Z}_1^{\max}} \quad (27)$$

Priority Selection Given the competition efficiencies η_{12} for objective Z_1 to Z_2 , and η_{21} vice versa, the regions are divided based on the advantage:

- If $(\eta_{12} - 1) > \delta$, Z_1 is advantageous; the Pareto-optimal solutions should be near the maximum value of Z_1 .
- If $(\eta_{21} - 1) > \delta$, Z_2 is advantageous; the Pareto-optimal solutions should be near the maximum value of Z_2 .
- If $\max(\eta_{12}\eta_{21}) - 1 < \delta$, the advantages of Z_1 and Z_2 are regarded as equivalent.

The solutions are evaluated against line l , which signifies a balance between Z_1 and Z_2 . The preferred regions are then defined by the following constraints:

- Region Ω_{12} is preferred when Z_1 is advantageous:

$$\begin{cases} Z_2 \geq k_{AB}Z_1 + b_{AB}, \\ Z_2 \leq k_l Z_1 + b_l, \end{cases} \quad (28)$$

- Region Ω_{21} is preferred when Z_2 is advantageous:

$$\begin{cases} Z_2 \geq k_{AB}Z_1 + b_{AB}, \\ Z_2 \geq k_l Z_1 + b_l, \end{cases} \quad (29)$$

- Region Ω'_{12} is preferred when Z_1 and Z_2 have comparable competitive advantages:

$$\begin{cases} Z_2 \geq k_{AB}Z_1 + b_{AB}, \\ -\tau_1 \leq Z_2 - k_l Z_1 - b_l \leq \tau_2. \end{cases} \quad (30)$$

Parameters k_{AB} , b_{AB} , k_l , b_l , τ_1 , and τ_2 are determined by the geometric relationships of the objectives in the decision space. Region Ω'_{12} is particularly delineated by two lines parallel to l passing through points D_1 and D_2 , which define the division ratios AD_1 and AD_2 :

$$\tau_1 = \frac{\delta d_{AB}}{2(2 - \delta) \sin \theta}, \quad \tau_2 = \frac{\delta d_{AB}}{2(2 + \delta) \sin \theta}, \quad (31)$$

where θ is the angle subtended by line AB and the line of objectives at their extreme points.

By applying these constraints, the Pareto front is effectively narrowed to the most competitive solutions as per the established criteria of competition efficiency.

2.3.3. Final Shrinking Region (FSR)

The intersection results in a Final Shrinking Region (Ω_{FSR}) that combines the insights from both KPITU and SMQC strategies:

$$\Omega_{FSR} = \Omega_{KPITU} \cap \Omega_{SMQC}. \quad (32)$$

This hybrid region yields a selection of solutions that are vetted for both their relative trade-off value and their peak objective performance. It offers a smaller, more manageable set of choices that integrate the KPITU region, which are important for balancing among solution sets, and the SMQC region which is significant for the competitive efficiency of extremal points. As a result, the FSR provides a flexible and pragmatic way to select solutions, facilitating a more robust and informed decision-making process on the Pareto front.

The essence of PFSS lies in its ability to intersect the KPITU region—known for underscoring critical trade-offs between solutions—with the SMQC region—recognized for pinpointing trade-offs at the extremities of objective values. The intersection of these regions forms a strategic Final Shrinking Region (Ω_{FSR}):

Through this hybridization, PFSS filters solutions that demonstrate exceptional balance in trade-offs as well as peak performance in objectives, resulting in a refined subset of the Pareto front. The strategy delineates a condensed yet comprehensive decision space, balancing the nuanced precision of knee points against the broader efficacy portrayed by the competition efficiency. Therefore, the PFSS offers a sophisticated and adaptive method for identifying an elite set of solutions, enhancing decision-making capabilities for multi-objective optimization.

2.4. The proposed framework

The proposed ensemble learning framework follows these steps:

- **Step 1: Data Preprocessing:** SVMd is used to decompose the data, with the highest-frequency component identified as noise and removed. HI is then applied to remove outliers. Finally, the ReliefF algorithm selects the most important features, reducing dimensionality and preventing overfitting.

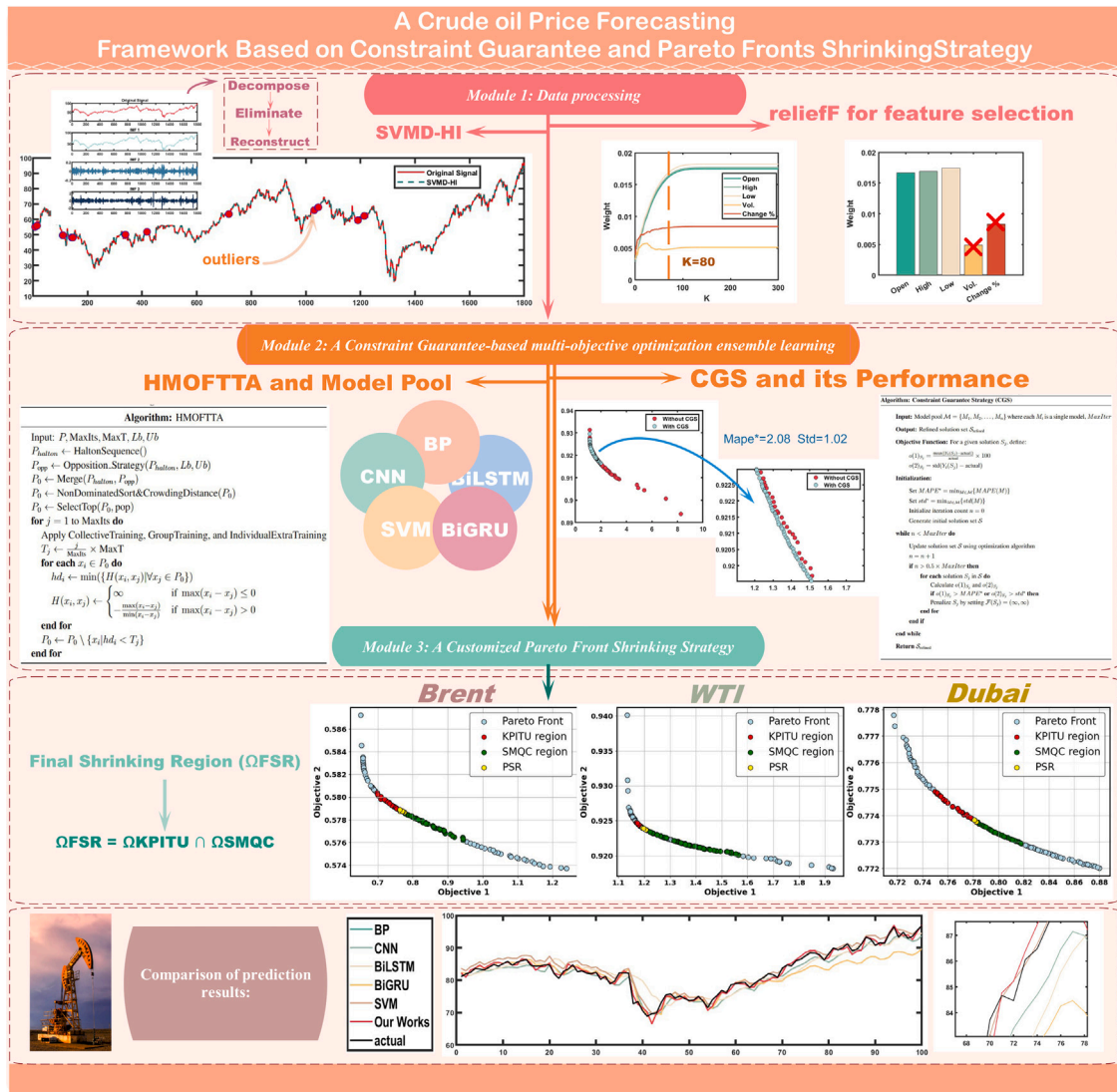


Fig. 1. Flowchart of the Complete Process. This figure illustrates the data flow across the three modules of the proposed framework: data preprocessing, ensemble learning, and the customized Pareto Frontier Shrinking Strategy Module.

- Step 2: Model Prediction:** Each model in the pool generates individual predictions. These include both deep learning and shallow models, resulting in a set of predictions for each data point.
- Step 3: Multi-objective Optimization Ensemble Learning with Constraint Guarantee:** A multi-objective optimization function is set to minimize MAPE and standard deviation (STD) on the validation set. In the latter half of the optimization process, constraints are applied to filter out solutions that do not outperform the best model in the pool in terms of accuracy and stability.
- Step 4: Optimization with HMOFTTA:** The enhanced multi-objective optimization algorithm, HMOFTTA, is employed to solve the optimization problem, determining the best solution that balances accuracy and stability.
- Step 5: Model Weight Determination using PFSS:** The Pareto Frontier Shrinking Strategy (PFSS) is applied to narrow down the solution set on the Pareto front. The final solution set is selected, and the optimal weight for each model is determined based on its contribution to the ensemble.
- Step 6: Linear Weighted Ensemble:** The final model prediction is obtained by applying the model weights to each model's predictions, then calculating a linear weighted average to combine the results.

Fig. 1 illustrates the technical process flow of the entire framework.

3. Data selection, parameter design and evaluation indexes

3.1. Data selection

This study analyzes crude oil price dynamics using data from the three most globally recognized benchmarks: Brent, West Texas Intermediate (WTI), and Dubai. Data was collected over 1800 trading days, with one data point corresponding to each day, starting from March 15, 2015. The dataset, which includes features like opening price, closing price (the prediction target), maximum and minimum values, trading volume, and date, is divided into a training set (first 1500 days), a validation set (next 200 days) and a test set (last 100 days), to facilitate comprehensive model training, validation, and testing. The dataset is publicly available on (Investing.com). The objective of this study is to predict the closing price. Fig. 2 presents the line charts and probability distributions of the closing prices for each dataset. Table 5 shows the statistical features of each dataset.

Table 6 show the original features and the selected features after applying the ReliefF algorithm for feature selection. In the table, "None" indicates that a feature is missing in the dataset. Fig. 3 illustrates the implementation details of the ReliefF algorithm. The ReliefF algorithm

Table 5
Basic characteristics of crude oil price.

	Max	Min	Mean	Std.	Var.	Data Amount	Training Sst	Validation Set	Testing Set
Brent	96.84	19.33	58.35	13.70	187.43				
WTI	84.65	-37.63	53.36	12.32	151.82	1800	1500	200	100
Dubai	122.53	19.07	57.69	15.69	246.12				

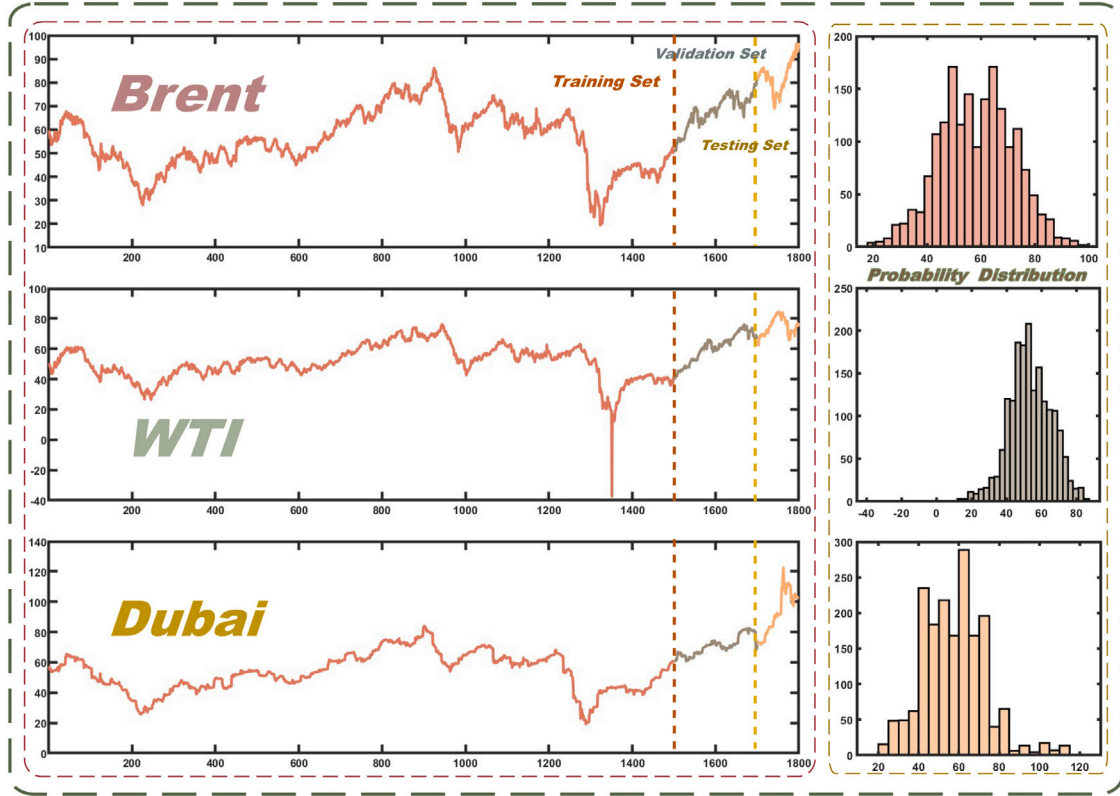


Fig. 2. Basic characteristics of crude oil price data (x-axis: Time points, y-axis: \$/bbl). The left panel shows the time-series plot of the dataset, while the right panel presents the probability histogram of the crude oil prices.

Table 6
Features selected by the ReliefF algorithm.

Dataset	Open	High	Low	Volume	Change %	Date(fixed)
Brent	✓	✓	✓	✗	✓	✓
WTI	✓	✓	✓	✗	✓	✓
Dubai	✓	✓	✓	None	✗	✓

assigns weights to each feature based on its importance, retaining significant features and discarding irrelevant ones to avoid overfitting. The optimal value of the parameter 'k' is determined by adjusting it from 1 to 300 and selecting the value at the inflection point, balancing predictive accuracy and feature relevance.

3.2. Parameter design

This section introduces the parameters of core models used in this research. Table 7 outlines the hyperparameters associated with these methods, while Table 8 details the internal parameters of three deep learning models and two shallow models. All parameters are set in their standard forms. The table outlines the search ranges for some key

hyperparameters, with an asterisk (*) indicating the values adopted in this study.

3.3. Evaluation indexes

We employ a suite of evaluation metrics to effectively gauge the performance of models. These metrics are crucial for understanding prediction accuracy and guiding model refinement. The following is a summary of the principal metrics used in our analysis.

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \frac{|y_i - \hat{y}_i|}{y_i} \times 100\% \quad (33)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (34)$$

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (35)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (36)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (37)$$

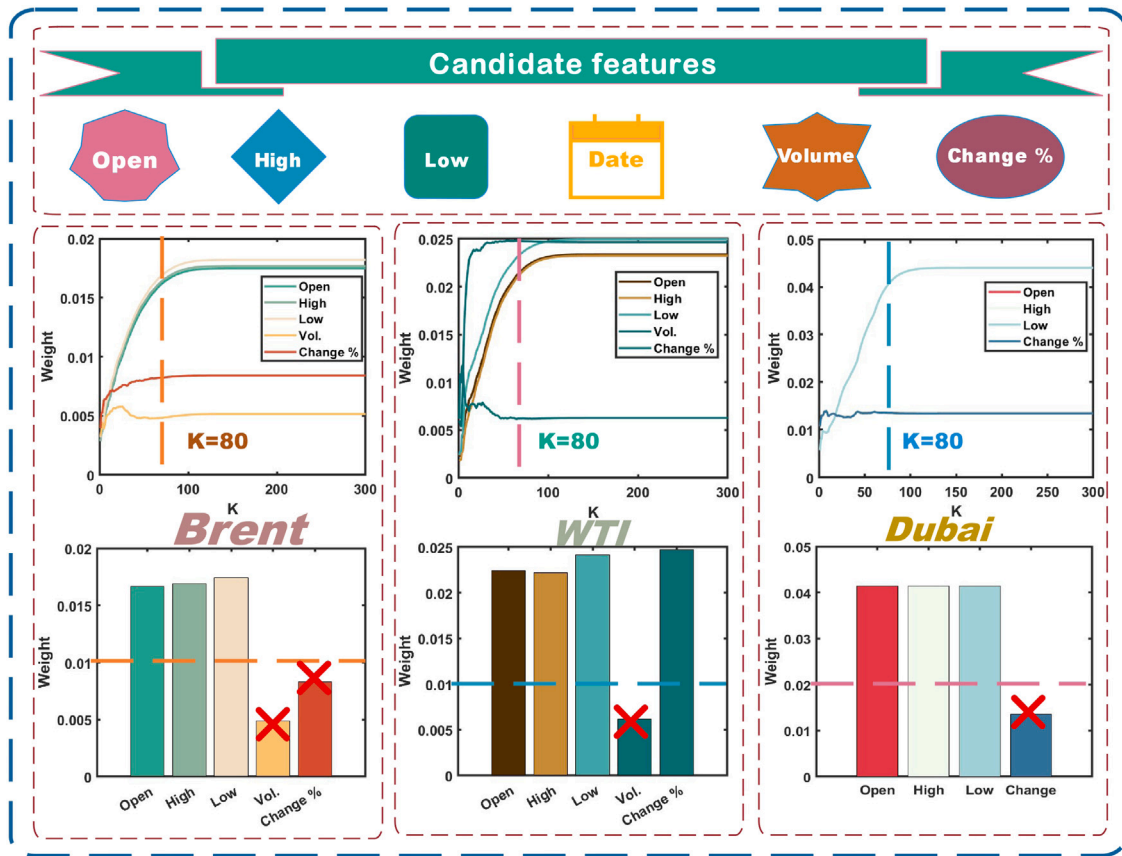


Fig. 3. The performance of ReliefF. The top panel shows the candidate features from the dataset. The middle panel presents the weight curves of different features as the value of k changes, with the optimal k value selected at the inflection point. The bottom panel illustrates the feature weights corresponding to the selected k value, where features are filtered based on a predefined threshold.

Table 7
Hyper-parameters of our framework.

Model	Parameters	Values
SVMD	MaxAlpha	10
	Tau	0.5
	Tol	$1e-3$
	Stopc	4
	Init omega	1
HI	Window size	6
	Threshold	3
ReliefF	K (number of nearest neighbors)	80
	Distance metric	Euclidean
Model Pool	Time step	100
	Batch size	64
	Max epoch	200
	Learning rate	0.001
	Drop output rate	0.25
	Output size	1
KPITU	K (the amount of knee point)	20
SMQC	delta	0.5
HMOFTTA	Pop	100
	Archive	100
	MaxIter	1000
	P_{siduy}	0.2
	P_{comm}	0.2
	P_{error}	0.001

Here, y_i is the actual value and $\hat{y}_i$ is the predicted value for the i^{th} instance, $\bar{y}$ and $\bar{\hat{y}}$ represent the mean of actual and predicted values respectively, n denotes the total number of instances, and $\sigma_{\hat{y}}$ is the standard deviation of predicted values.

3.4. Operation environment

The fundamental operating environment for the proposed crude oil price forecasting framework consists of an Intel Core i7-13620H processor and an RTX4050 graphics card. Data preprocessing and multi-objective optimization were conducted using Matlab 2022a, while the deep learning structures were built using PyTorch within the PyCharm integrated development environment.

4. Numerical validation

4.1. Performance comparison across models

In this experiment, we compare our proposed model with three state-of-the-art (SOTA) models from the general time series forecasting domain and three advanced models specifically designed for crude oil price prediction. Table 9 summarizes the performance of these models across three different datasets: Brent, WTI, and Dubai crude oil prices. The general SOTA models include iTransformer [46], Informer [47], and DLinear [48], while the advanced models for crude oil price prediction include the Jaynes Weight Hybrid (JWH) model [28], VMD-PSR-CNN-LSTM [24], and C_V-BiGRU_A_C [26]. All parameter settings are based on the original text, or obtained by grid search if the original text is not available. Below, we provide a brief description of these models. Table 10 shows the comparison results of Experiment 1.

Although general SOTA models such as iTransformer, Informer, and DLinear have demonstrated strong performance in general time series forecasting tasks, they fall short when applied to specific domains like crude oil price prediction [49]. For instance, models specifically

Table 8
Inter-parameters of deep learning shallow models.

	Model	Parameters	Values
Deep Learning	CNN	Layer Number	[2*, 3, 4]
		Kernel size	[3*, 5, 7]
		Padding	1
		Pooling Kernel size	[3*, 5, 7]
		Pooling Stride	[2*, 3, 4]
	BiLSTM	LSTM Layer Number	[1, 2*, 3]
		LSTM Hidden size	[64, 128*, 256]
Shallow Model	SVM	GRU Layer Number	[1, 2*, 3]
		GRU Hidden size	[64, 128*, 256]
		C	[50, 100*, 200]
	MLP	Epsilon	[0.001, 0.005*, 0.01]
		Gamma	[0.005*, 0.01]
		Layer Number	[2, 3, 4*]
		Hidden size	[128, 256*, 512]

Table 9
Overview of the models used for comparison in Experiment 1.

Model	Description
iTransformer	A transformer model for time series forecasting that inverts dimensions (time steps as features, features as sequences) to enhance global dependency capture.
Informer	A transformer-based model that introduces an efficient attention mechanism to reduce the complexity of long time series predictions, suitable for large datasets.
DLinear	A linear model that uses deep learning to combine different linear models in a time series context, offering a good balance between accuracy and complexity.
JWH	A hybrid model that incorporates Shannon information entropy and time-varying weights to improve prediction accuracy in volatile environments like crude oil price forecasting.
VMD-PSR-CNN-LSTM	A hybrid prediction model that first denoises the crude oil price series using variational mode decomposition (VMD) and phase space reconstruction (PSR), followed by convolutional neural networks (CNN) and long short-term memory (LSTM) networks for prediction.
C_V_BiGRU_A_C	A combined model using secondary decomposition and a BiGRU-Attention-CNN network, which has shown superior performance in forecasting crude oil futures prices and volatility.

designed for crude oil price forecasting, like VMD-PSR-CNN-LSTM, outperform the general models. In the Brent dataset, VMD-PSR-CNN-LSTM achieves a MAPE of 1.4858, which is significantly lower than iTransformer's 1.8195 and Informer's 1.8346. This indicates that domain-specific models are better at capturing the unique characteristics and volatility of the oil market.

Furthermore, DLinear, while a general model, still performs better than iTransformer-based models. In the same dataset, DLinear achieves a MAPE of 1.7409, outperforming iTransformer's 1.8195. This suggests that DLinear, despite being a general model, is more robust and stable compared to iTransformer, especially when dealing with the complexity of crude oil price data.

Our model outperforms all the general time series forecasting models, such as iTransformer, Informer, and DLinear, across all datasets. For instance, in the Brent dataset, our model achieves the lowest MAPE of 0.9062, significantly outperforming iTransformer (1.8195), Informer (1.8346), and DLinear (1.7409). Additionally, our model achieves the highest R2 of 0.9796, which is considerably higher than iTransformer (0.9050), Informer (0.9063), and DLinear (0.9172). These results highlight the superiority of our model in predicting crude oil prices compared to general-purpose models designed for general time series tasks.

Furthermore, when compared to domain-specific crude oil price prediction models like JWH, VMD-PSR-CNN-LSTM, and C_V_BiGRU_A_C, our model still outperforms in both accuracy and robustness. For example, in the WTI dataset, our model achieves a MAPE of 1.6149,

which is lower than JWH (1.9355) and VMD-PSR-CNN-LSTM (1.6651). In the Dubai dataset, our model achieves the lowest MAPE of 1.2284, outperforming JWH (1.8735) and VMD-PSR-CNN-LSTM (1.4834). Additionally, in terms of R2, our model again leads with 0.9198 for WTI and 0.9843 for Dubai, surpassing both JWH and VMD-PSR-CNN-LSTM.

Remark 1. The numerical validation demonstrates that our proposed model outperforms both state-of-the-art models from the general time series forecasting domain and domain-specific models for crude oil price prediction. Our framework's ability to achieve superior accuracy and robustness across various datasets highlights its potential as a reliable tool for forecasting crude oil prices.

4.2. Necessity of data denoising and feature engineering

In this section, we utilize the BiGRU model to perform experiments across three datasets, highlighting the importance of our integrated SVM-D-HI-ReliefF preprocessing strategy. To ensure fair comparison, all other components of the framework were held constant. We then compared the full preprocessing strategy with approaches that involve only feature engineering, only data denoising, or no preprocessing at all. The results of these experiments are presented in Table 11.

The results in Table 11 highlight a critical insight: both data denoising and feature engineering play crucial roles in enhancing model performance, but the degree of improvement varies across datasets. Notably, applying only the ReliefF method to the Brent dataset results in a substantial enhancement, with MAPE improving by 58.76%, MSE decreasing by 76.48%, and R² increasing by 42.52%. This indicates that feature engineering significantly aids in improving predictive accuracy by emphasizing the most relevant features. Similarly, when SVM-D-HI is applied for data denoising alone, all metrics show marked improvement, and the enhancement is even greater than that achieved through feature engineering alone.

However, the key takeaway here is the synergistic effect when both techniques are combined. The combination of SVM-D, HI, and ReliefF leads to the most significant performance improvements across all datasets, although the increase in performance is not always as large as when each method is applied individually. This demonstrates that the integrated approach, which combines data denoising and feature engineering, is consistently superior to applying either technique in isolation.

For the WTI and Dubai datasets, similar improvements are observed, with both data denoising and feature engineering contributing to enhanced performance. Combining these methods results in further gains, albeit to a lesser degree. This suggests that while the combination of these methods yields improvements, the magnitude of the benefit depends on the specific characteristics of the dataset.

Table 10
Prediction results of three sets of data in Experiment 1.

Dataset	Model	MAPE	MAE	MSE	RMSE	R ²
Brent	iTransformer	1.8195	1.4770	4.0657	2.0163	0.9050
	Informer	1.8346	1.5152	4.0104	2.0026	0.9063
	DLinear	1.7409	1.4313	3.5457	1.8830	0.9172
	JWH	1.8772	1.5546	3.7840	1.9453	0.9116
	VMD-PSR-CNN-LSTM	1.4858	1.2063	2.9199	1.7088	0.9318
	C_V_BiGRU_A_C	1.1801	1.0110	1.4901	1.2207	0.9652
	Our work	0.9062	0.7383	0.8716	0.9336	0.9796
WTI	iTransformer	2.1894	1.6083	4.3784	2.0925	0.8695
	Informer	2.2554	1.7005	4.6732	2.1618	0.8607
	DLinear	1.6642	1.2132	2.8259	1.6811	0.9157
	JWH	1.9355	1.4345	3.2412	1.8003	0.9034
	VMD-PSR-CNN-LSTM	1.6651	1.2154	2.9264	1.7107	0.9128
	C_V_BiGRU_A_C	1.6821	1.2505	2.7668	1.6634	0.9175
	Our work	1.6149	1.1974	2.6893	1.6399	0.9198
Dubai	iTransformer	1.9505	1.8953	8.6529	2.9416	0.9559
	Informer	3.3392	3.2185	18.4767	4.2985	0.9058
	DLinear	1.8907	1.8185	7.7345	2.7811	0.9606
	JWH	1.8735	1.8161	7.4688	2.7329	0.9619
	VMD-PSR-CNN-LSTM	1.4834	1.4615	6.6908	2.5867	0.9659
	C_V_BiGRU_A_C	1.5698	1.5808	4.5378	2.1302	0.9769
	Our work	1.2284	1.1743	3.0811	1.7553	0.9843

Table 11
Prediction results of three datasets in Experiment 1.

	Method	MAPE	MAE	MSE	RMSE	R ²
Brent	Original-BiGRU	4.0452	3.4310	15.3086	3.9126	0.6423
	Original-ReliefF-BiGRU	1.6683	1.4020	3.6006	1.8975	0.9159
	SVMD-BiGRU	2.1591	1.7476	4.6600	2.1587	0.8911
	HI-BiGRU	1.9276	1.6025	4.2087	2.0515	0.9017
	SVMD-HI-BiGRU	1.7457	1.4934	3.5878	1.8941	0.9162
	SVMD-HI-ReliefF-BiGRU	1.5613	1.2713	2.7295	1.6521	0.9362
WTI	Original-BiGRU	2.8111	2.1556	6.8149	2.6105	0.7968
	Original-ReliefF-BiGRU	1.9177	1.4301	3.5052	1.8722	0.8955
	SVMD-BiGRU	2.6893	2.0143	4.5633	2.1362	0.8639
	HI-BiGRU	2.0525	1.4773	4.5028	2.1220	0.8658
	SVMD-HI-BiGRU	2.0992	1.6252	4.1303	2.0323	0.8769
	SVMD-HI-ReliefF-BiGRU	1.6916	1.2342	2.7780	1.6667	0.9172
Dubai	Original-BiGRU	3.0052	3.0072	16.4487	4.0557	0.9161
	Original-ReliefF-BiGRU	2.8273	2.7947	15.2357	3.9033	0.9223
	SVMD-BiGRU	2.4143	2.2445	7.7062	2.7760	0.9607
	HI-BiGRU	1.8371	1.7673	7.9928	2.8272	0.9592
	SVMD-HI-BiGRU	2.2574	2.1102	6.8690	2.6209	0.9650
	SVMD-HI-ReliefF-BiGRU	1.4454	1.4380	5.6484	2.3766	0.9712

Remark 2. The experiments demonstrate that both data denoising and feature engineering individually contribute to improving predictive performance across various datasets. However, the degree of improvement depends on the dataset characteristics. Notably, the combination of these preprocessing methods consistently results in superior predictive outcomes compared to using either technique alone, reinforcing the importance of integrated data preprocessing in enhancing model performance.

4.3. Superiority of ensemble learning

The primary objective of Experiment 3 was to demonstrate the superiority of our proposed forecasting framework for crude oil prices compared to individual model predictions. By maintaining consistency across other modules in the framework, we aimed to validate two critical aspects. First, we sought to confirm that the integration of ensemble learning within our framework significantly enhances prediction performance. Second, we aimed to show that even when single models adopt the techniques embedded in our framework, they still do not achieve the same level of accuracy and stability, thereby highlighting our framework's ability to outperform individual predictive

methods without the need for parameter adjustments. The experimental data presented in Table 12 provide a detailed comparison between the performance of single models and our framework across 11 key indicators, setting the stage for a comprehensive analysis of the results.

The results in Table 12 clearly demonstrate that our framework exhibits superior performance across all three datasets. The incorporation of CGS ensures that the ensemble's predictive accuracy consistently exceeds the best-performing individual model in the pool, while also providing a more robust overall prediction by leveraging multiple models.

A critical observation is the lack of robustness among individual models when applied to different datasets. The performance of these models varies significantly, highlighting a key limitation of current single AI methods. For instance, in the experimental results, the MLP performed well on the Brent dataset, achieving an R² of 0.9230 and an MSE of 3.2940. However, on the Dubai dataset, its performance declined considerably, with an R² of only 0.8475 and an MSE of 29.9049. In contrast, our ensemble framework effectively corrects this variability by dynamically adapting to the contributions of different models across datasets. It assigns greater weights to models that perform well on a given dataset while penalizing those that detract from

Table 12
Prediction results of three sets of data in Experiment 3.

	Method	MAPE	MAE	MSE	RMSE	R ²
Brent	SVMD-HI-MLP	1.7660	1.4617	3.2940	1.8149	0.9230
	SVMD-HI-CNN	2.3110	1.8491	6.8283	2.6131	0.8405
	SVMD-HI-BiLSTM	2.5014	2.1648	9.1801	3.0299	0.7855
	SVMD-HI-BiGRU	1.5613	1.2713	2.7295	1.6521	0.9362
	SVMD-HI-SVM	2.3850	1.9885	4.6481	2.1559	0.8914
	Our work	0.9062	0.7383	0.8716	0.9336	0.9796
WTI	SVMD-HI-MLP	2.4236	1.8024	4.3583	2.0876	0.8701
	SVMD-HI-CNN	4.0591	2.9905	12.1834	3.4905	0.6368
	SVMD-HI-BiLSTM	3.1052	2.3546	7.9316	2.8163	0.7635
	SVMD-HI-BiGRU	1.6916	1.2342	2.7780	1.6667	0.9172
	SVMD-HI-SVM	3.9928	3.0509	11.4369	3.3819	0.6590
	Our work	1.6149	1.1974	2.6893	1.6399	0.9198
Dubai	SVMD-HI-MLP	5.2295	4.9561	29.9049	5.4685	0.8475
	SVMD-HI-CNN	3.4171	3.3428	26.1691	5.1156	0.8666
	SVMD-HI-BiLSTM	1.5693	1.4987	5.4221	2.3285	0.9724
	SVMD-HI-BiGRU	1.9662	1.9232	7.2530	2.6931	0.9630
	SVMD-HI-SVM	1.7155	1.6825	5.1676	2.2732	0.9736
	Our work	1.2284	1.1743	3.0811	1.7553	0.9843

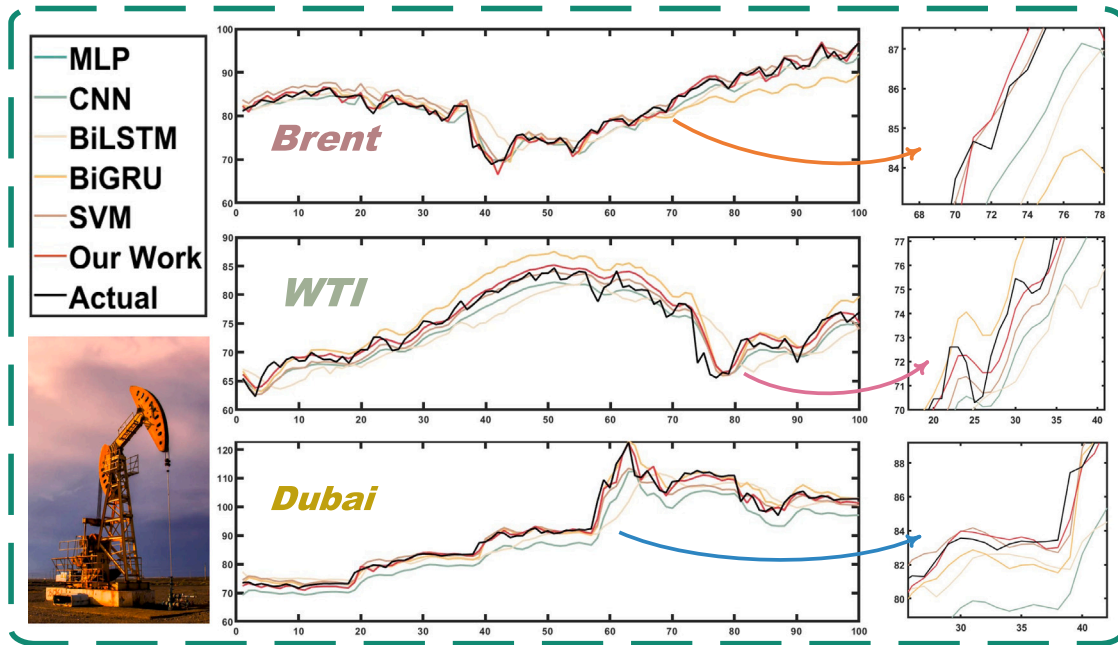


Fig. 4. The performance of Our work vs. Single Model. (x-axis: Time points, y-axis: \$/bbl). This figure demonstrates the predictive performance of Our work compared to individual AI models on test datasets from different datasets. Our work exhibits the best prediction accuracy.

the ensemble's overall performance, even assigning negative weights where necessary. Additionally, the integration of CGS ensures that our ensemble framework always outperforms the best model in the pool, regardless of the dataset.

Fig. 4 visually illustrates the performance on the test set, showing how closely the red line, representing our framework's predictions, aligns with the black line, which indicates the actual values. This close alignment highlights the accuracy of our framework in capturing the true trends of the data.

Remark 3. Experiment 3 demonstrates that our ensemble framework not only enhances the predictive accuracy of individual AI methods but also ensures robust and stable predictions, especially in scenarios where decision-makers lack prior knowledge about which model is best suited for a specific dataset. The incorporation of CGS further strengthens our framework, ensuring that it consistently outperforms the best individual model in the pool across any dataset. This improvement enhances

decision-making efficiency and reduces the decision-making cost for stakeholders.

4.4. The advantages of constraint guarantee strategy and the customized pareto front shrinking strategy

To rigorously validate the advantages of the Constraint Guarantee Strategy (CGS) and Pareto Front Shrinking Strategy (PFSS) within our framework, we conducted Experiment 4, which compared several selection strategies on the Pareto front. These strategies included random selection, KPITU knee point selection, random selection with CGS, and our integrated CGS-PFSS approach. All other components of the framework remained unchanged. The results from this experiment are summarized in [Table 13](#).

First, the integration of CGS into the framework results in notable enhancements in prediction accuracy across all datasets. Compared to baseline methods such as random selection and KPITU knee point selection, the models incorporating CGS consistently achieve

Table 13
Prediction results of three datasets in Experiment 4.

	Method	MAPE	MAE	MSE	RMSE	R ²
Brent	SVMD-HI-MOEL-RAND	1.4236	1.1633	1.9232	1.3868	0.9551
	SVMD-HI-MOEL-KPITU	1.6803	1.3771	2.5548	1.5984	0.9403
	SVMD-HI-MOEL-CGS	1.1357	0.9248	1.3303	1.1534	0.9689
	SVMD-HI-MOEL-CGS-PFSS	0.9062	0.7383	0.8716	0.9336	0.9796
WTI	SVMD-HI-MOEL-RAND	1.9600	1.4503	3.4924	1.8688	0.8959
	SVMD-HI-MOEL-KPITU	2.2844	1.6947	4.4518	2.1099	0.8673
	SVMD-HI-MOEL-CGS	1.7936	1.3324	3.1795	1.7831	0.9052
	SVMD-HI-MOEL-CGS-PFSS	1.6149	1.1974	2.6893	1.6399	0.9198
Dubai	SVMD-HI-MOEL-RAND	1.3377	1.2773	3.3646	1.8343	0.9828
	SVMD-HI-MOEL-KPITU	1.2106	1.1539	3.1131	1.7644	0.9841
	SVMD-HI-MOEL-CGS	1.3116	1.2506	3.2799	1.8110	0.9833
	SVMD-HI-MOEL-CGS-PFSS	1.2284	1.1743	3.0811	1.7553	0.9843

Table 14
Reduction in Inferior Solutions at the Start of CGS.

Dataset	Inferior Solutions Reduced	Percentage Reduced
Brent	28	≈ 33%
WTI	31	≈ 33%
Dubai	72	> 50%

better performance. For example, on the Brent dataset, the MAPE decreases from 1.4236 (random selection) and 1.6803 (KPITU) to 1.1357 (CGS), demonstrating that CGS improves the ensemble's performance by reducing the error. Similar improvements are observed in the WTI and Dubai datasets, where CGS reduces MAPE and other performance metrics compared to the baseline methods.

Additionally, we tracked the reduction in inferior solutions within the archive when CGS was initiated. As shown in Table 14, initiating CGS led to a reduction of approximately one-third of the inferior solutions in the archive for both the Brent and WTI datasets, and for the Dubai dataset, the reduction exceeded half of the inferior solutions, which fail to guarantee that the ensemble model's performance will surpass that of individual models. By focusing the optimization process on high-quality solutions, CGS ensures better ensemble performance, thereby improving prediction accuracy. Fig. 5 presents a comparison between the use of the CGS and its absence, clearly demonstrating that employing CGS ensures that the optimization algorithm can search within a more effective area. These areas guarantee that the predictive performance of the ensemble learning significantly surpasses that of any single model. Furthermore, the magnified section in Fig. 5 illustrates that the Pareto optimal solutions obtained with the CGS dominate those without CGS. It further demonstrates that CGS not only ensures an enhancement in predictive performance but also improves the quality of the Pareto frontier.

Second, while CGS successfully filters out inferior solutions that lack practical application value, the multi-objective optimization process still generates a set of Pareto-optimal solutions. PFSS helps decision-makers objectively select solutions from the Pareto front that guarantee predictive accuracy. As shown in Table 13, PFSS further improves prediction accuracy. For example, in the Brent dataset, the MAPE improves from 1.4236 (with random selection) to 0.9062 (with CGS-PFSS), and the MAE decreases from 1.1633 to 0.7383. This demonstrates that PFSS effectively reduces the solution space and ensures that the decision-making process leads to more accurate predictions, ultimately improving performance.

Further, Table 15 displays the number of Pareto-optimal solutions obtained from each process. Initially, the Pareto front contains 100 solutions. PFSS refines this set by taking the intersection of the solutions filtered by KPITU and SMQC, reducing the number of solutions in the final solution set (FSR). Specifically, PFSS reduces the initial 100 solutions to 5 in the Brent dataset, to 4 in the WTI dataset, and to 2 in the Dubai dataset. This reduction directly impacts the efficiency

Table 15
The number of Pareto optimal solutions obtained from each process.

	KPITU	SMQC	FSR
Brent	16	39	5
WTI	11	33	4
Dubai	9	25	2

of model selection, ensuring that decision-makers are presented with a smaller, more relevant set of solutions while maintaining or improving predictive performance. Fig. 6 illustrates the performance of PFSS, showing how it distills the optimal solutions from the Pareto front. By focusing on the trade-offs along the Pareto front, PFSS reduces the solution space, making the decision-making process more efficient. This not only saves decision-makers time but also ensures that the selected model configuration is robust across different datasets, further demonstrating the utility of PFSS in guiding the decision-making process.

Remark 4. The results from Experiment 4 demonstrate that the integration of CGS and PFSS not only improves prediction accuracy but also enhances the decision-making process for decision-makers working with limited prior information. The reduction in inferior solutions achieved by CGS and the efficient solution selection provided by PFSS streamline the optimization process, ensuring better overall predictive outcomes. These strategies significantly reduce decision-making costs and increase efficiency, ensuring the best possible outcomes are achieved with minimal effort.

4.5. The efficiency of HMOFTTA

As previously discussed, our optimization problem necessitates the use of heuristic optimization algorithms. To demonstrate the superior solving capabilities of our selected MOFTTA and the improved HMOFTTA, we conducted a comparison under consistent conditions against well-known algorithms such as NSGA2 and MOGWO. Based on the experimental results presented in Table 16, two key insights can be drawn regarding the performance of the optimization algorithms.

First, the MOFTTA algorithm demonstrates superior performance compared to other well-known multi-objective optimization algorithms, such as NSGA2 and MOGWO. Across all three datasets, MOFTTA consistently achieves lower MAPE, MAE, and MSE values, along with higher R² scores, indicating its stronger capability in optimizing the predictive accuracy of our ensemble learning framework. For example, in the Brent dataset, MOFTTA achieved a MAPE of 1.0076, which is a notable improvement over NSGA2 (1.1489) and MOGWO (1.1612). This suggests that MOFTTA is better suited for solving the complex optimization problems inherent in our framework, particularly when balancing multiple objectives. Second, our multi-strategy improved HMOFTTA further outperforms the original MOFTTA in all key metrics. The results show that HMOFTTA achieves the best performance across all

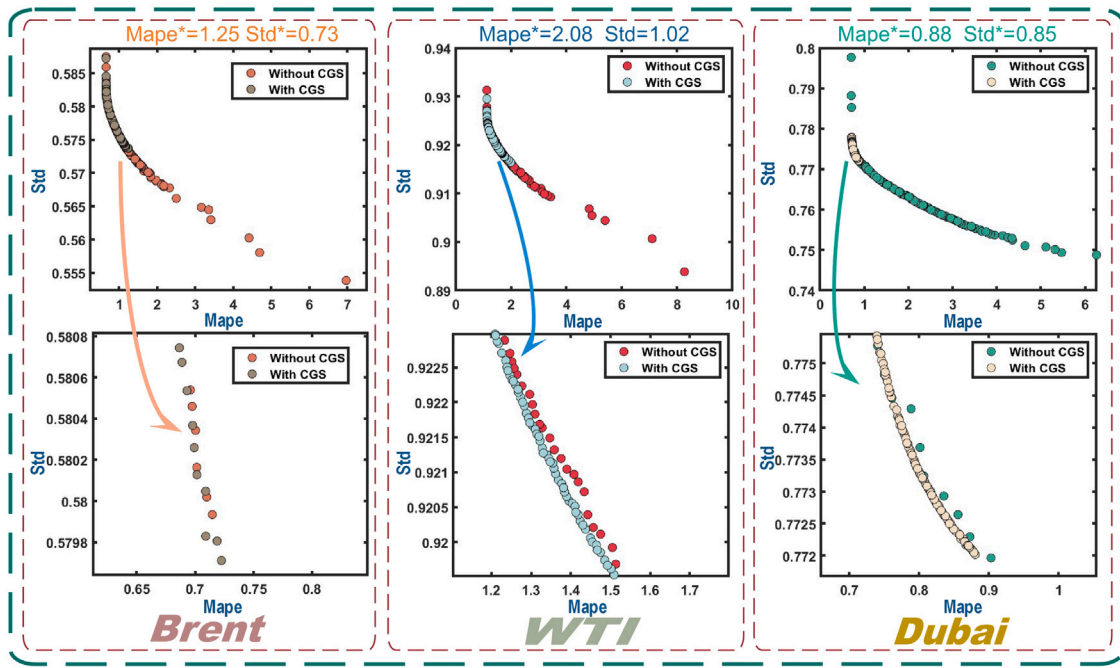


Fig. 5. The performance of CGS. The upper part of the figure compares the Pareto fronts obtained with and without using CGS. The lower part provides a zoomed-in view, illustrating that the solutions obtained with CGS dominate those without CGS, demonstrating better performance.

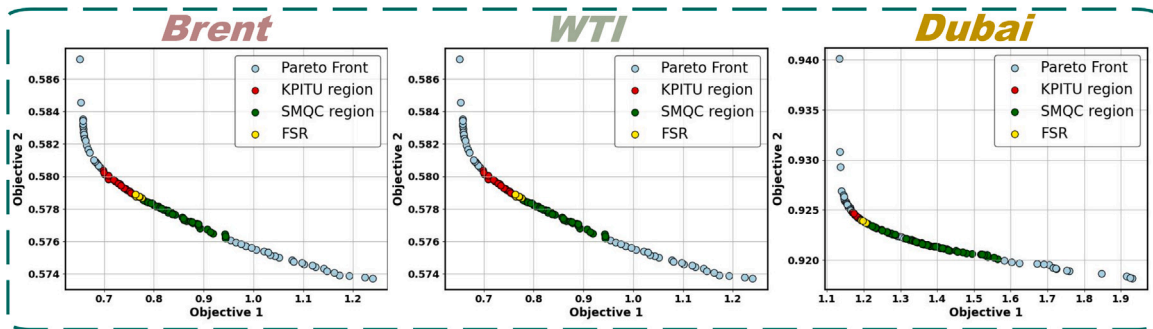


Fig. 6. The performance of PFSS. This figure shows how PFSS refines the Pareto front by focusing on the intersection of the KPITU and SMQC regions. The yellow points represent the intersection, which is used to obtain the Final Solution Set (FSR).

Table 16

Prediction results of three datasets in Experiment 5.

	Method	MAPE	MAE	MSE	RMSE	R ²
Brent	NSGA2-CGS-PFSS	1.1489	0.9371	1.3613	1.1668	0.9682
	MOGWO-CGS-PFSS	1.1612	0.9507	1.3805	1.1749	0.9677
	MOFTTA-CGS-PFSS	1.0076	0.8184	1.0593	1.0292	0.9752
	HMOFTTA-CGS-PFSS	0.9062	0.7383	0.8716	0.9336	0.9796
WTI	NSGA2-CGS-PFSS	1.7927	1.3362	3.2244	1.7957	0.9039
	MOGWO-CGS-PFSS	1.7491	1.2985	3.0369	1.7427	0.9095
	MOFTTA-CGS-PFSS	1.6733	1.2376	2.7572	1.6605	0.9178
	HMOFTTA-CGS-PFSS	1.6149	1.1974	2.6893	1.6399	0.9198
Dubai	NSGA2-CGS-PFSS	1.4657	1.4260	3.9274	1.9818	0.9800
	MOGWO-CGS-PFSS	1.4845	1.4629	4.1556	2.0385	0.9788
	MOFTTA-CGS-PFSS	1.3400	1.2721	3.3643	1.8342	0.9828
	HMOFTTA-CGS-PFSS	1.2284	1.1743	3.0811	1.7553	0.9843

datasets, with even lower error values and higher R² scores than those obtained with MOFTTA. For instance, in the Brent dataset, HMOFTTA

achieves the lowest MAPE of 0.9062, outperforming MOFTTA (1.0076), MOGWO (1.1612), and NSGA2 (1.1489). Similarly, in the WTI dataset,

Table 17
The results of ZDT in Discussion 1.

	Algorithm	IGD	SPREAD	HV
ZDT1	MOGWO	1.947E-03	1.314E+00	8.555E-01
	NSGA2	7.290E-04	4.749E-01	8.697E-01
	MOFTTA	5.700E-04	3.795E-01	8.709E-01
	HMOFTTA	4.890E-04	2.786E-01	8.715E-01
ZDT2	MOGWO	5.484E-03	1.340E+00	5.037E-01
	NSGA2	7.590E-04	5.099E-01	5.368E-01
	MOFTTA	6.000E-04	3.498E-01	5.364E-01
	HMOFTTA	5.490E-04	3.615E-01	5.378E-01
ZDT4	MOGWO	3.784E-03	1.170E+00	8.439E-01
	NSGA2	7.710E-04	4.865E-01	8.697E-01
	MOFTTA	6.130E-04	4.278E-01	8.707E-01
	HMOFTTA	4.880E-04	2.820E-01	8.714E-01
ZDT6	MOGWO	2.027E-02	9.490E-01	3.979E-01
	NSGA2	7.790E-04	1.140E+00	4.307E-01
	MOFTTA	4.360E-04	3.202E-01	4.322E-01
	HMOFTTA	3.870E-04	3.054E-01	4.322E-01

HMOFTTA achieves a MAPE of 1.6149, lower than MOFTTA's 1.6733, and in the Dubai dataset, HMOFTTA's MAPE of 1.2284 is significantly better than MOFTTA's 1.3400. This improvement highlights that the modifications introduced in HMOFTTA make it more effective for our specific optimization problem, ensuring more accurate and reliable predictions.

Remark 5. While MOFTTA already provides a strong foundation for optimizing predictive accuracy within our framework, the multi-strategy improved HMOFTTA proves to be even more effective, confirming its suitability and superiority for solving the complex multi-objective optimization problems posed by our ensemble learning approach. The improvements in prediction accuracy across all datasets further validate the advantages of HMOFTTA, showcasing its robustness and efficiency in enhancing our model's performance.

5. Discussions

5.1. How does HMOFTTA perform on the various ZDT test functions?

To further demonstrate the performance of MOFTTA and our multi-strategy improved HMOFTTA on multi-objective optimization problems, we evaluated their performance against MOGWO and NSGA2 using the ZDT1, ZDT2, ZDT4, and ZDT6 test functions. Each algorithm was assessed using three metrics: IGD, Spread, and HV. The results revealed that HMOFTTA consistently exhibited superior performance. Table 17 displays the detailed comparative data, while Fig. 7 shows the Pareto fronts of different optimization algorithms on various test functions.

First, the quantitative assessment using IGD, Spread, and HV highlighted HMOFTTA's strengths. SPREAD, which measures the uniformity of solutions, showed that HMOFTTA achieved the smallest values on three of the four test functions, indicating highly uniform Pareto fronts. For example, in ZDT1, HMOFTTA achieved a Spread value of 0.2786, significantly better than MOFTTA (0.3795), MOGWO (1.3140), and NSGA2 (0.4749). Even on ZDT2, where it did not achieve the smallest value, the difference was minimal, showcasing its consistent performance. HV and IGD further illustrated HMOFTTA's superiority in convergence and diversity. HMOFTTA consistently achieved the highest HV and the lowest IGD values across most test functions, indicating better coverage of the objective space and closer approximation to the true Pareto front compared to other algorithms. For example, in ZDT4, HMOFTTA achieved the highest HV (0.8714) and the lowest IGD (4.880E-04), outperforming the other algorithms.

The qualitative evaluation through Pareto front visualizations in Fig. 7 reinforced these findings. HMOFTTA generated more uniformly distributed and accurate Pareto fronts, closely aligning with the true

Pareto front, unlike the other algorithms. For instance, in ZDT1, HMOFTTA produced a Pareto front that was more compact and closely matched the true front, whereas the Pareto fronts from MOGWO and NSGA2 were more spread out.

In conclusion, HMOFTTA outperforms classical multi-objective optimization algorithms and the original MOFTTA in convergence, uniformity, and diversity. The improvements in HMOFTTA make it a robust and effective tool for solving complex optimization problems, solidifying its role within our framework.

5.2. Is our framework practical for real-time trading?

To assess the practical applicability of our framework in real-time trading, we compare it against three advanced state-of-the-art (SOTA) models from general domains and three advanced models from the crude oil price forecasting domain, as described in Experiment 1. We also conduct a comparison using a variant of our framework where the Pareto Frontier Shrinking Strategy (PFSS) is removed. Our experiment is conducted over a 100-day test period on three datasets: Brent, WTI, and Dubai. A trading strategy is designed based on these models' predictions.

The trading strategy works as follows: if the predicted change in futures prices for the next day compared to the current day is greater than 5%, the model buys; if the predicted change is less than -5%, the model sells all holdings; otherwise, the model holds. The initial capital for all models is set to \$10,000.

Fig. 8 shows the change in capital over time for each of the models, along with the strategies adopted at each time point (1 for buy, 0 for hold, and -1 for sell). Table 18 presents the final capital for each model across the three datasets. As shown in the results, our framework achieves the highest final capital in all three datasets. Specifically, in Brent, WTI, and Dubai datasets, the final capital for our framework is \$19,345.61, \$15,905.15, and \$19,905.75 respectively, outperforming both the other models and our framework without PFSS. This demonstrates that our framework's ability to adapt to different datasets, coupled with the integration of PFSS, leads to steady and substantial growth in capital. The model without PFSS, while still profitable, lags behind in all three datasets, highlighting the added value and practical significance of PFSS in real-world trading.

In conclusion, our framework provides superior performance in real-time trading, offering higher returns and improved stability compared to other models. The incorporation of PFSS significantly enhances the overall performance, demonstrating its value in optimizing decision-making and maximizing economic benefits in financial applications.

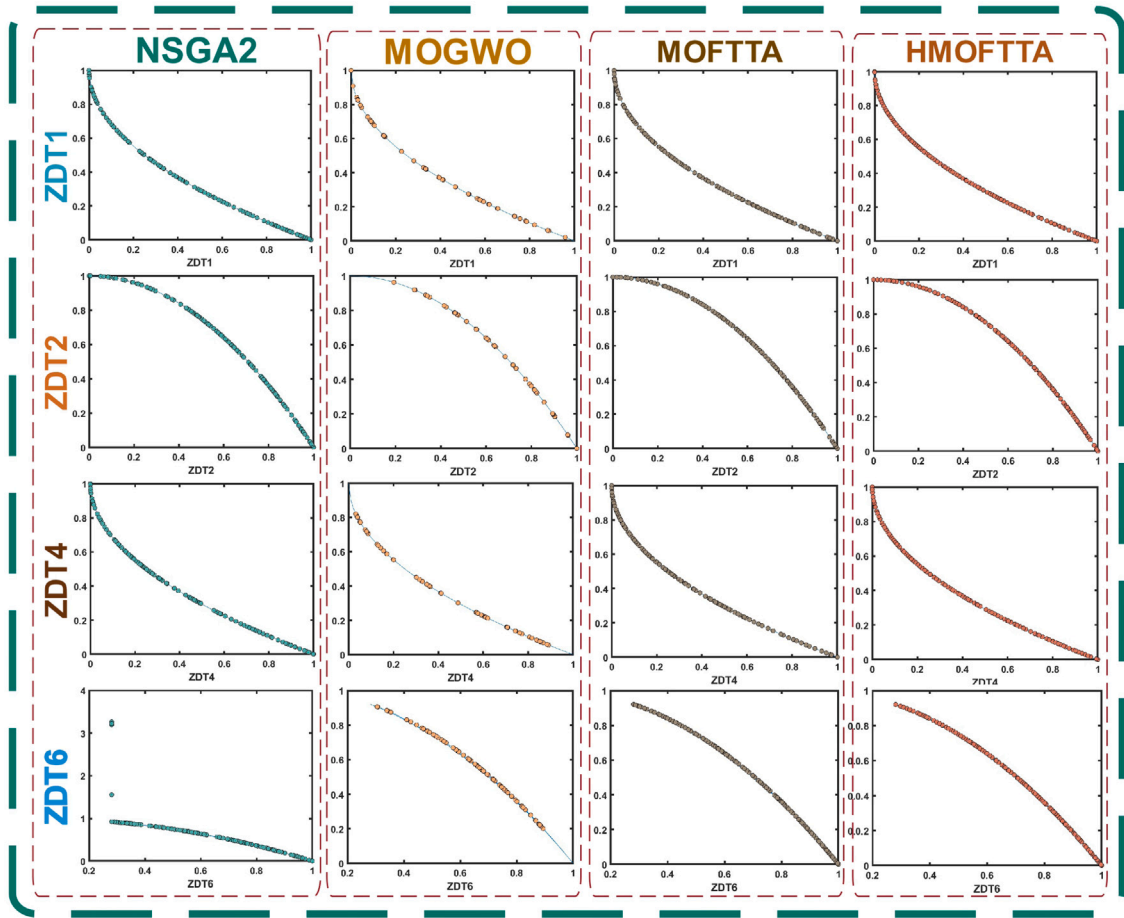


Fig. 7. The performance of Pareto fronts on various test functions. This figure shows the Pareto fronts obtained by different multi-objective optimization algorithms on various test functions. HMOFTTA generates a Pareto front that is closer to the true Pareto front and more evenly distributed compared to the other algorithms.

Table 18

Capital Accumulation for Each Model over 100 Days.

Dataset	iTransformer	Informer	DLinear	JWH
Brent	9836.2006	12 480.4508	11 035.4365	10 000.0000
WTI	11 333.3737	11 550.0715	13 726.2314	11 104.1920
Dubai	11 608.4619	13 621.2582	10 712.0321	12 742.7652
Dataset	VMD-PSR-CNN-LSTM	C-V-BiGRU-A-C	Our work w/o PFSS	Our work
Brent	13 105.6756	16 247.1916	15 426.9871	19 345.6118
WTI	10 812.4109	15 504.5225	14 950.8725	15 905.1539
Dubai	13 581.5922	18 831.8529	17 725.7670	19 905.7451

5.3. Is the computational burden of the proposed framework appropriate for the engineering application?

In practical engineering applications, it is crucial to assess the computational complexity of a model to understand its feasibility for real-world deployment. A comprehensive analysis of computational burden helps determine whether the model can efficiently handle large datasets and provide timely results in real-time systems. In this context, runtime, the number of floating-point operations (FLOPs), and the number of parameters serve as key indicators of both time and space complexity [50]. Table 19 presents the computation times, FLOPs, and parameter counts for each model in the proposed framework.

From the table, we observe that the proposed framework operates with satisfactory efficiency despite being composed of multiple sub-models. The overall runtime of the framework is 195.32 s, which is under four minutes—an efficient duration for a complex model.

While integrating multiple sub-models typically leads to longer computational times due to the increased complexity, the proposed framework maintains an acceptable runtime, making it feasible for practical applications where time is of the essence.

When evaluating the space complexity, the FLOPs and parameter counts for each model offer insight into the computational load. The most computationally demanding sub-model is the MLP-CNN-BiLSTM-BiGRU-SVM, with 219.61 million FLOPs and 1.83 million parameters. This model, while intricate, plays a key role in the framework's overall performance. The proposed framework itself has 391.97 million FLOPs and 1.83 million parameters, which are considered manageable for most real-time applications.

In conclusion, the computational efficiency of the proposed framework, as demonstrated by its runtime, FLOPs, and parameter counts, is well-suited for engineering applications. The framework effectively

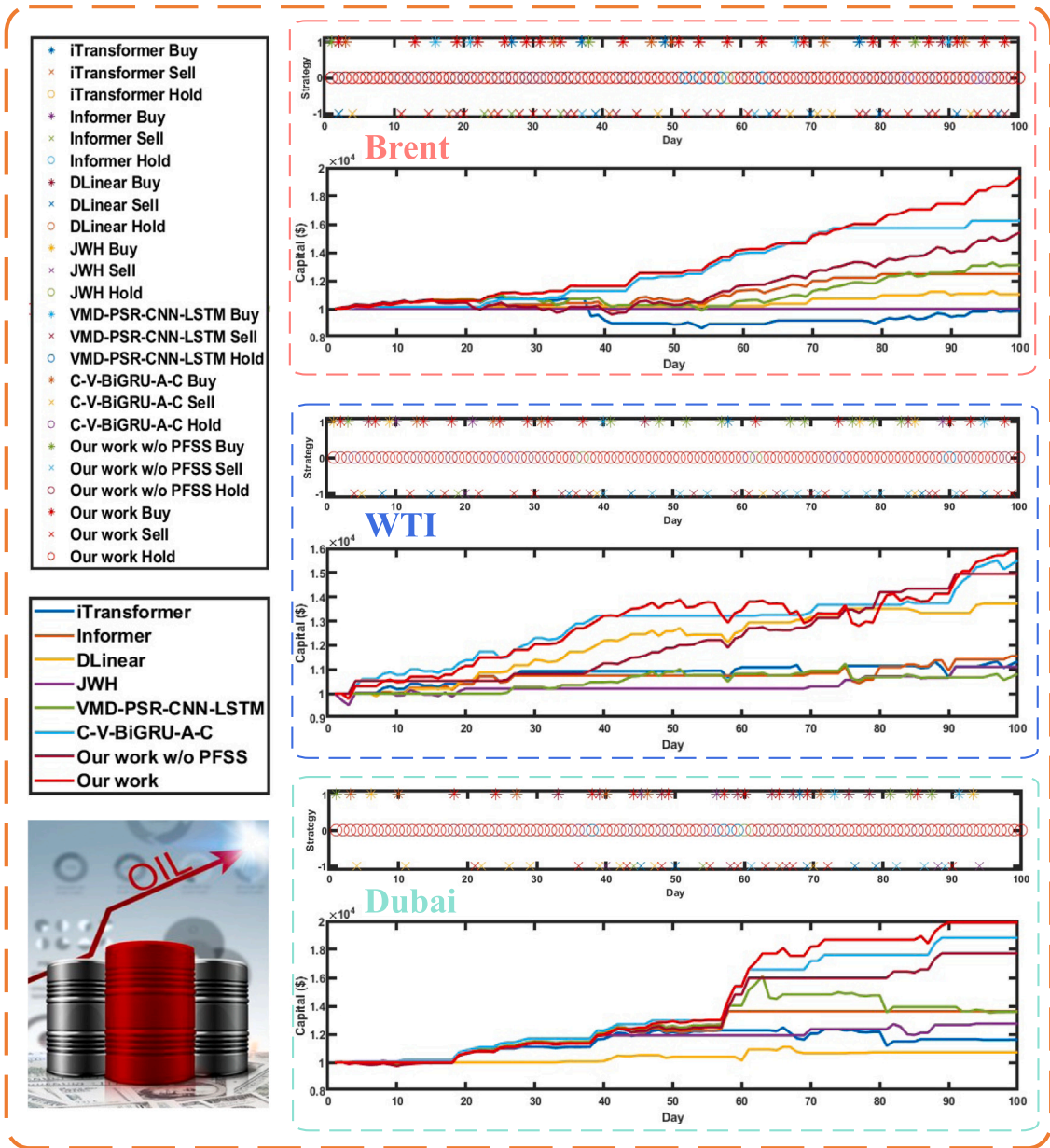


Fig. 8. Real-time trading capital change over time. This figure shows the capital accumulation for each model over a 100-day test period on different datasets (Brent, WTI, and Dubai). The strategies adopted at each time point are indicated: 1 for buying, 0 for holding, and -1 for selling. The change in capital reflects the performance of each model's trading strategy, with our framework showing the highest growth across all datasets.

Table 19
Running time, FLOPs, and Parameters of each model.

Model	Computation time (s)	FLOPs	Params
SVMD-HI-ReliefF	29.27	4.12 M	$\times$
MLP-CNN-BILSTM-BIGRU-SVM	110.44	219.61 M	1.83 M
Ensemble Learning Based on CGS	54.62	167.42 M	$\times$
PFSS	0.98	0.82 M	$\times$
Proposed framework	195.32	391.97 M	1.83 M

handles multiple sub-models while maintaining a manageable computational load, with runtime and parameter counts being acceptable for practical use cases. The results suggest that the proposed framework can be deployed in real-time environments, providing accurate and timely predictions without excessive computational costs. Such efficiency makes it a promising tool for practical engineering applications requiring large-scale forecasting.

6. Conclusion

This study introduced a novel framework aimed at improving the accuracy, robustness, and applicability of crude oil price forecasts. By integrating the Constraint Guarantee Strategy (CGS) and a customized Pareto Front Shrinking Strategy (PFSS) within an ensemble learning framework, we have developed a reliable tool to address the challenges associated with forecasting in the volatile crude oil market.

The proposed framework is composed of three core components: data preprocessing, ensemble learning with CGS, and the PFSS. The data preprocessing module, utilizing SVMD-HI-ReliefF, effectively processes the input data, enhancing model training and reducing the risk of overfitting. The ensemble learning module, strengthened by CGS, dynamically adjusts model weights, penalizing underperforming solutions to ensure that the ensemble consistently outperforms individual models. The PFSS simplifies the decision-making process by facilitating the selection of Pareto front solutions, especially in situations where prior knowledge is limited.

The long-term impact of this research lies in its potential to transform forecasting practices in not only the crude oil market but also in other financial markets where forecasting accuracy is crucial. By improving both predictive accuracy and decision-making capabilities, the framework could aid policymakers, investors, and stakeholders in navigating complex and volatile market conditions.

While the framework demonstrates notable improvements, there are limitations. First, the current ensemble learning method assigns fixed weights to the individual models, but model weights may exhibit time heterogeneity. Different time periods could require different patterns, which may need distinct weights to reflect changes in market behavior. Second, the current approach uses linear integration of model outputs. Although this significantly improves forecasting accuracy, nonlinear methods may offer theoretical advantages in capturing complex relationships, potentially enhancing model performance further.

Future research could explore the following directions:

- **Time-Varying Weights:** Investigating the use of time-varying weights to allow the model to adapt to different market regimes, ensuring that it reflects the evolving nature of the crude oil market over time.
- **Nonlinear Ensemble Methods:** Exploring nonlinear integration methods, which could potentially improve the forecasting accuracy by capturing more complex relationships between the model predictions.
- **Incorporating Market Sentiment:** Introducing market sentiment factors, such as news sentiment and social media sentiment, to enhance the model's ability to respond to external influences on crude oil price movements.

CRedit authorship contribution statement

Yujie Chen: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Investigation, Formal analysis, Data curation, Conceptualization. **Zhirui Tian:** Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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